Small-Sample (*t-tools*) Confidence Intervals

Recall: If we are sampling from a *Normal* population with known variance then a confidence interval for μ is

$$\bar{Y} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$ (valid for any sample size).

Further, if the variance is unknown but we have a large sample size, then an approximate confidence interval for μ is

$$\bar{Y} \pm z_{\alpha/2} \frac{S}{\sqrt{n}}.$$

What if the variance is unknown and the sample size is small?

➔From definition 7.2,

$$T = \frac{\bar{Y} - \mu}{S / \sqrt{n}} \sim t_{n-1}$$ can be used as a pivot.

Small-Sample (*t-tools*) Confidence Intervals

Target Parameter: μ

1. If we are sampling from a Normal population with unknown variance then we have the following pivotal quantity,

$$T = \frac{\bar{Y} - \mu}{S / \sqrt{n}} \sim t_{n-1}.$$

2. For $T \sim t_{n-1}$, $P(-t_{n-1,\alpha/2} \le T \le t_{n-1,\alpha/2}) = 1 - \alpha.$

3. Then, $P(-t_{n-1,\alpha/2} \le T \le t_{n-1,\alpha/2}) = P\left(\bar{Y} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}} \le \theta \le \bar{Y} + t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}\right) = 1 - \alpha.$

$\therefore$ The two-sided $(1-\alpha)100\%\ C.I.$ for μ is $\bar{Y} \pm t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}$.

Note: If the sample size is large this interval still applies and will be approximately the same as $\bar{Y} \pm z_{\alpha/2}\frac{S}{\sqrt{n}}$.

<u>Recall Example 6:</u> The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounce.

a) Suppose a sample of size ***n*** = 9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces? Answer the question by looking at confidence intervals for the true mean.

b) Repeat part (a) supposing that the standard deviation was 6.

c) What is the minimum sample size, ***n***, so that the sample mean will be within 0.3 ounces of the population mean, 90% of the time (upon repeated sampling)?

d) Repeat parts (a) – (c) supposing that the standard deviation was estimated.

Confidence Intervals for Variances

Suppose we are sampling from **Normal populations**.

Here we will develop confidence intervals for:

1) σ^2 (a single population variance)

2) σ_1^2/σ_2^2 (the ratio of two population variances)

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed $Normal(\mu_Y, \sigma_Y^2)$ random variables.

Example 4d) Define confidence interval(s) for σ_Y^2 as a function of S^2.

1. Pivot: Consider $g(S^2, \sigma^2) = \dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

Suppose you have independent samples from different populations.

- n_1 observations from $X \sim N(\mu_1, \sigma_1^2)$ and
- n_2 observations from $Y \sim N(\mu_2, \sigma_2^2)$

Example 4 f) Define a confidence interval to compare population variances.

1. Pivot: Consider $g((S_1^2, S_2^2), (\sigma_1^2, \sigma_2^2)) = \dfrac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F_{n_2-1}^{n_1-1}$